

OPAL 1: growth by recorded label

The effect of nickel on growth is undetermined from this export.

Study design at a glance

The export supports a structured description—not confirmed dose groups

Recorded label	Plants	Genotypes	Status in this analysis
0	60	12	Categorical; not confirmed control
50	59	12	Categorical; meaning/unit unconfirmed
100	59	12	Categorical; meaning/unit unconfirmed
200	59	12	Categorical; meaning/unit unconfirmed
500	60	12	Categorical; meaning/unit unconfirmed
1000	60	12	Categorical; meaning/unit unconfirmed

DESIGN UNITS

359 plants

GENOTYPES

12

CELLS

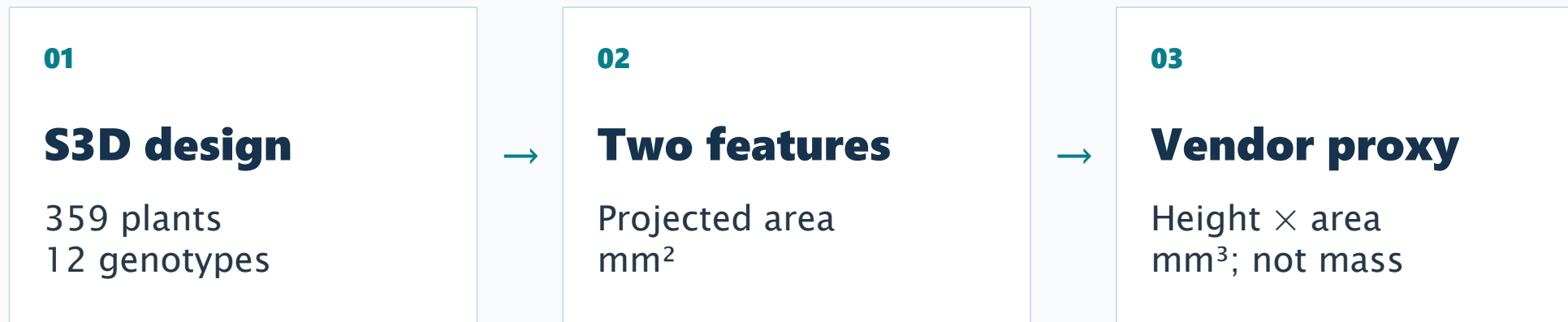
72 × 4–5 plants

ROUNDS

22 dates

What was measured—and what it is not

22 rounds • 2026-04-08 to 2026-05-07 • timestamps have no UTC offset

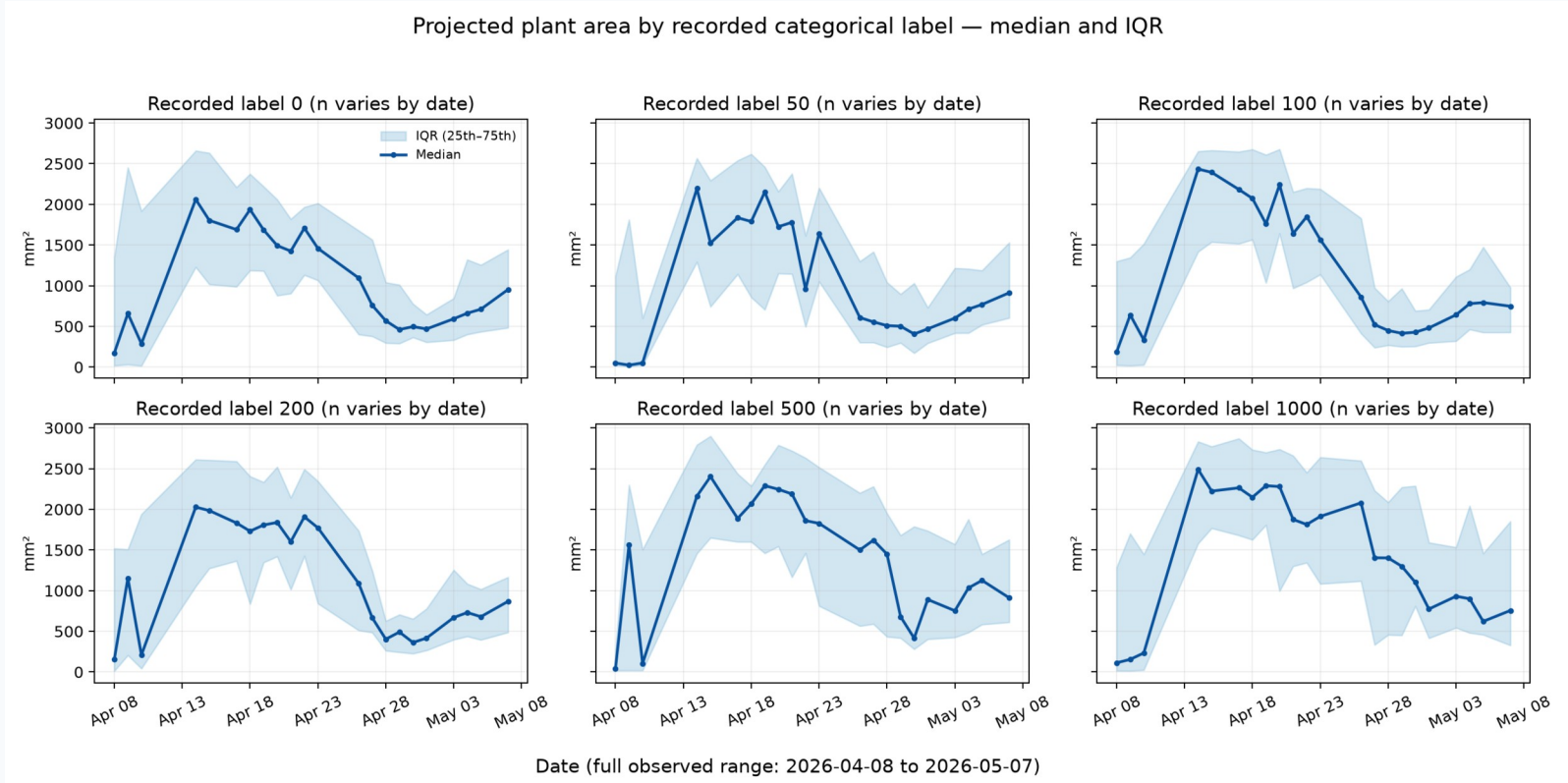


Processing: plant × date × trait summaries; repeated same-day scans averaged before trajectories/slopes.

7,794 plant–date–trait rows after selection. Feature QC flags are empty; no rows excluded. Five exact-zero area values and source outliers retained.

Projected area: trajectories vary across labels

Median with IQR (25th–75th percentile); full observed date range retained



READOUT

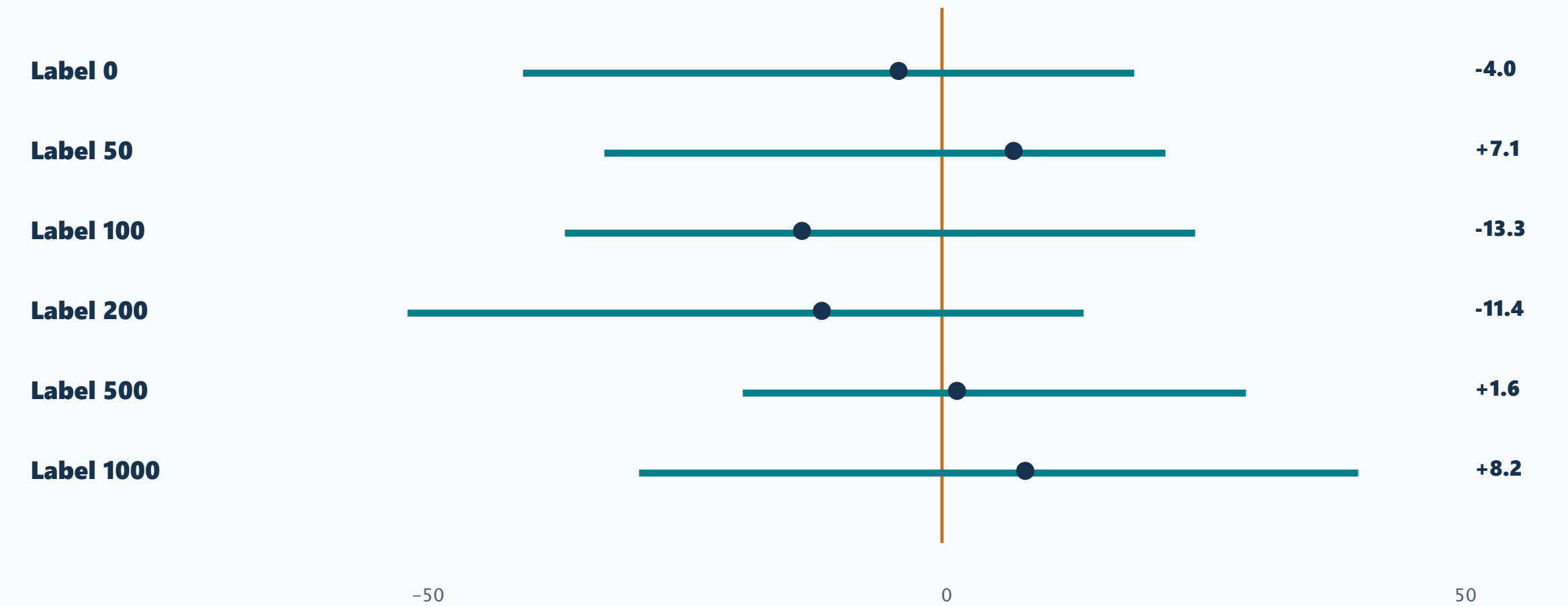
No consistent penalty pattern is evident in these descriptive trajectories.

Wide within-label spread; sample counts vary by date. IQR is not a confidence interval.

labels remain categorical

Per-plant area slopes: all IQRs cross zero

Median and IQR of plant-level linear slopes (mm²/day); plant is replication unit

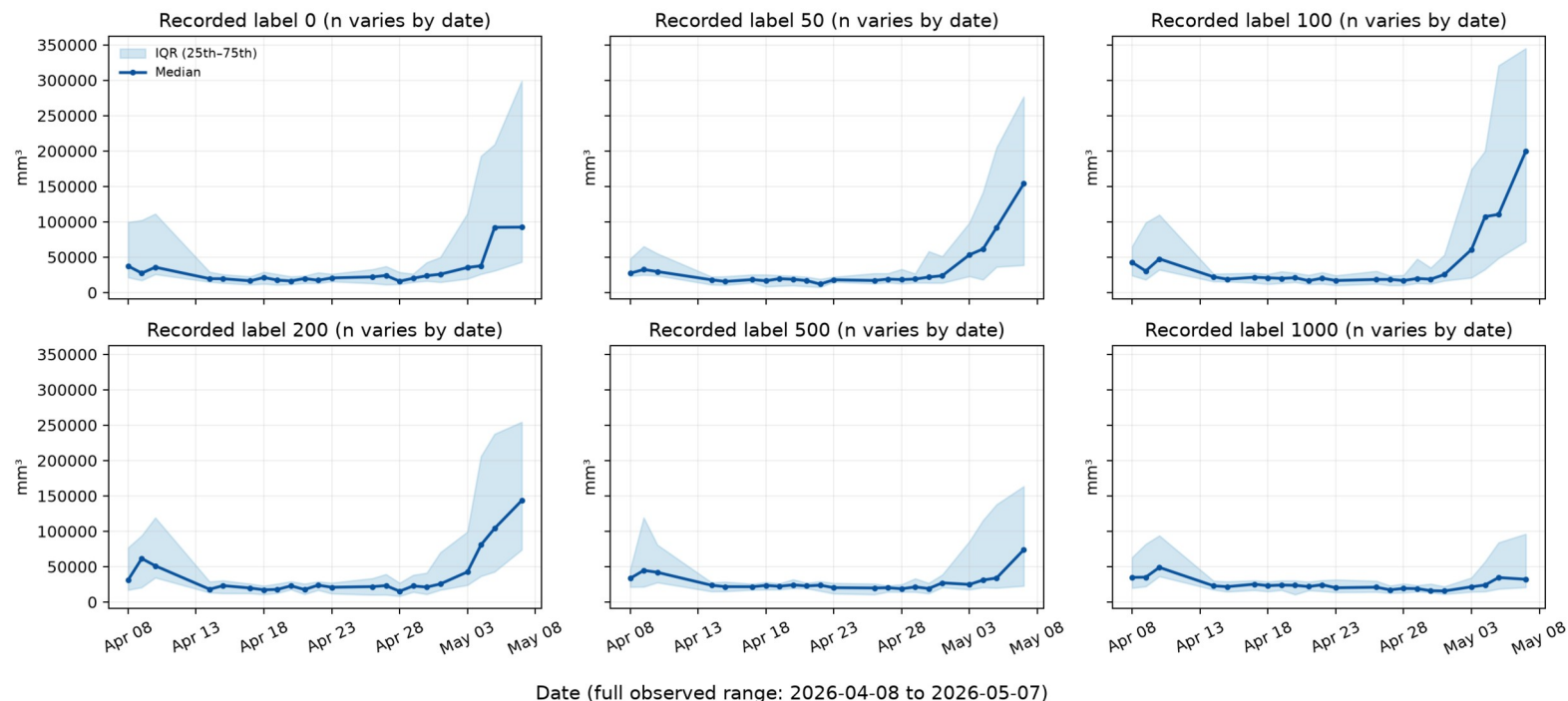


Ranges show 25th–75th percentile across plants—not confidence intervals or treatment–effect estimates.

The vendor proxy breaks sharply early in the series

Digital biomass is vendor height $\times$ area (mm³), not measured plant mass

Digital-biomass proxy by recorded categorical label — median and IQR



OBSERVED DISCORDANCE

Proxy median
44,553 → 20,820.85 mm³

Area median
174 → 2,164.5 mm²

April 10 → April 14. Cause is not established by exported QC/status.

Paired dates tell a discordant measurement story

Overall date medians above; paired within-plant changes below (n=108 matched plants)

PROXY

Digital-biomass proxy

44,553 mm³ → 20,820.85 mm³

–33,240.65 mm³ median paired change

AREA

Projected area

174 mm² → 2,164.5 mm²

+889.5 mm² median paired change

Interpretation: proxy and area move in opposite directions. This is a quality/measurement question—not evidence of biological biomass loss.

Separate what is observed from what remains unknown

Use the evidence to target follow-up; do not fill metadata gaps with assumptions

SUPPORTED BY SAVED EXPORT

- 359 plants; 12 genotypes; six labels
- 22 rounds; 7,794 rows after selection
- No QC exclusions; five area zeros retained
- Area slope IQR spans zero for every label

NOT ESTABLISHED

- Which labels represent nickel
- Dose unit or concentration convention
- Whether any label is the control
- Cause of April 10–14 proxy discontinuity

Do not make a nickel growth claim yet.

Biological effect = UNDETERMINED from this export

01

Verify treatment mapping

Owner confirms identity, control, units

02

Resolve proxy discontinuity

APPL/S3D review of acquisition and analysis

03

Then estimate response

Genotype-aware repeated-measures analysis

Decision gate: proceed to biological interpretation only after label semantics and proxy quality are resolved.

Owner questions that unblock the next analysis

Discussion / assign follow-up

TREATMENT

Which recorded labels map to nickel, control and what dose units?

MEASUREMENT

What changed between April 10 and April 14 in S3D acquisition or processing?

TIME

What timezone applies to timestamp strings; are cross-sensor joins valid?

Source basis and quality guardrails

Backup / provenance

ANALYSIS INPUTS

Reviewed meeting summary
Detailed exploratory report
Analysis metadata and saved summary tables
Existing projected-area and digital-proxy figures

INTERPRETATION LIMITS

IQR = 25th–75th percentile, not CI
Date coverage varies
Source zeros/outliers retained
Timestamps lack UTC offset
No causal or genotype-adjusted estimate

Experiment 67 • APPL-CORE export v6 • source export manifest SHA-256 recorded in analysis:
76bf90275f7a98392fc6d1363b46cc3a28142ebfd219c8705e19636604771757